Project 1

Lane-Emden Equation

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PROJECT ASSIGNMENT: NUMERICAL SOLUTIONS FOR POLYTROPIC STELLAR STRUCTURES (LANE-EMDEN EQUATION)

0.1 PROJECT OVERVIEW AND THEORETICAL OBJECTIVES

This project demands a rigorous numerical treatment of the internal structure of stars through the polytropic approximation. By assuming a power-law relationship between pressure and density ($P = K\rho^{1+1/n}$), the fundamental equations of stellar equilibrium—mass conservation and hydrostatic balance—are reduced to the dimensionless Lane-Emden equation. This project assignment focuses on the sophisticated transition from abstract dimensionless models to the derivation of physical stellar parameters.
The core objective is to solve the Lane-Emden equation for varying polytropic indices:

$$\frac{1}{\xi^2}\frac{d}{d\xi}\left(\xi^2\frac{d\theta}{d\xi}\right) = -\theta^n \tag{0.1}$$

where θ represents the dimensionless temperature (density) and ξ is the dimensionless radius. The solution is constrained by the following boundary conditions at the stellar center ($\xi = 0$):

$$\theta(0) = 1$$
$$\theta'(0) = 0$$

Your analysis will bridge the gap between numerical integration and the physical characterization of a star's density concentration and central pressure.

NUMERICAL METHODOLOGY: INTEGRATION AND DIFFERENTIATION

IMPLEMENTATION OF THE RUNGE-KUTTA 4 (RK4) SOLVER

To apply standard integration techniques, convert the second-order Lane-Emden ODE into a coupled system of two first-order ODEs. Let $y_1 = \theta$ and $y_2 = \frac{d\theta}{d\xi}$. The system is defined as:

$$\frac{dy_1}{d\xi} = y_2 \tag{0.2}$$

$$\frac{dy_2}{d\xi} = -y_1^n - \frac{2}{\xi}y_2 \tag{0.3}$$

Coordinate Singularity Handling: You must address the $2/\xi$ singularity at the center ($\xi = 0$). You are required to derive and submit a small-ξ Taylor expansion to initialize your integration. Use the following expansion for $\theta(\xi)$ near the center to determine your starting values at a small $\Delta\xi$:

$$\theta(\xi) \approx 1 - \frac{1}{6}\xi^2 + \frac{n}{120}\xi^4 \tag{0.4}$$

Verification via Finite Differences

To verify the gradients (y_2) calculated during the RK4 integration, you must implement finite difference formulas at selected midpoints of your integration range. Use the central difference formula for a 3-point stencil. Higher-order formulae interpolation methodology are encouraged for increased precision during the verification phase.

Root Finding and Interpolation for the Stellar Surface (ξ_1)

The stellar surface is defined by the first root ξ_1 where $\theta(\xi_1) = 0$. Since the integration step h will likely overshoot this root, you must perform high-order interpolation (Cubic Splines or Newton's Divided Difference) specifically on the final bracketed interval $[\xi_i, \xi_{i+1}]$ where the sign change occurs to determine the precise value of ξ_1 and the surface gradient $\theta'(\xi_1)$.

Computational Task Specifications

Polytropic Indices and Global Integration

Solve the Lane-Emden equation for the following specific indices:

- $n = 3.00$
- $n = 3.15$
- $n = 3.30$
- $n = 3.45$
- $n = 3.60$
- $n = 3.75$

Data Visualization Requirements

Generate a high-quality diagnostic plot showing $\theta(\xi)$ vs ξ for all requested n values on a single set of axes. Ensure the plot includes a legend and clearly illustrates the migration of the surface root ξ_1 as the polytropic index increases.

Derivation of Physical and Dimensionless Parameters

Calculated Parameter Set

For each index n, construct a table containing:

1. First root ξ_1.
2. Surface gradient $\theta_1' = (d\theta/d\xi)|_{\xi_1}$.
3. Density concentration ratio $\rho_c/\overline{\rho}$.

4. Dimensionless mass parameter: $-\xi_1^2(d\theta/d\xi)|_{\xi_1}$.
5. Constants ${}_0\omega_n$, N_n, and W_n.
6. Central pressure P_c in 10^{17} dynes/cm^2.

Mandatory Physics Formulas and Constants

Use the following physical constants and formulas as the absolute basis for conversion:

- $G = 6.674 \times 10^{-8} cm^3 g^{-1} s^{-2}$
- $M_\odot = 1.989 \times 10^{33} g$
- $R_\odot = 6.957 \times 10^{10} cm$

Density Relations:

$$\rho_m = \frac{M_\odot}{\frac{4}{3}\pi R_\odot^3} \tag{0.5}$$

$$\rho_c = -\left[\frac{\xi_1/3}{(d\theta/d\xi)|_{\xi_1}}\right]\rho_m \tag{0.6}$$

$$\tag{0.7}$$

Dimensionless Constants:

$${}_0\omega_n = -\xi_1^{(n+1)/(n-1)}\left(\frac{d\theta}{d\xi}\right)\Big|_{\xi_1} \tag{0.8}$$

$$N_n = \frac{1}{n+1}\left(\frac{4\pi}{{}_0\omega_n^{n-1}}\right)^{1/n} \tag{0.9}$$

$$W_n = \frac{1}{4\pi(n+1)\left[-\xi_1^2\left(\frac{d\theta}{d\xi}\right)\Big|_{\xi_1}\right]^2} \tag{0.10}$$

Central Pressure:

$$P_c = W_n\left(\frac{GM_\odot^2}{R_\odot^4}\right) \tag{0.11}$$

Comparative Analysis: Polytropes vs. Standard Solar Model

Data Integration

Load the external dataset solar_model.txt (the Christensen-Dalsgaard standard solar model).

Profile Comparison

Map your numerical density profiles (ρ vs $R/R_\odot$) against the solar model data. Use interpolation to align your numerical grid with the standard solar model's data points for an accurate point-to-point comparison.

Visual and Analytical Comparison

Produce a plot comparing the density profile of the n=3.0 polytrope (the Eddington Standard Model) with the Christensen-Dalsgaard model.
Discussion Prompt: Within your report, discuss the discrepancies in density concentration. Specifically, evaluate how the $n = 3$ model tends to underestimate the core density of the actual Sun and why the polytropic approximation becomes less accurate in the deep interior.

Submission Requirements and Evaluation Criteria

Code Standards

The source code must be generalized to handle any value of n and must include comprehensive comments explaining the integration logic and verification steps.

6.2 Technical Report Format

Submit your results in a structured document report including at least:

1. Generalized Source Code.
2. Calculated Parameter Table for all six values of n.
3. Diagnostic Plots: The global $\theta(\xi)$ profile and the comparative density plot ($n = 3.0$ vs. Solar Standard).
4. Discussion: A brief analytical summary of how the surface gradient $(d\theta/d\xi)|_{\xi_1}$ and the central density concentration $\rho_c/\overline{\rho}$ evolve as the index n increases.